双末端闭式循环 · 公开教学示例 · 计算报告

OpenHydroNet 0.1.0-preview.5 | 2026-10-29T11:56:30.774878+00:00

状态：计算通过（converged）；算法：sparse-head-flow-newton-trf-v2

物理输入 SHA-256：f6e62dbcd139f0ad090e2fa44056deb4eb3a0fa9bb4957c7f9183f04237de623

假设和验证

水温	20 ℃
密度	998.2336 kg/m³
运动黏度	1.004e-06 m²/s
质量残差	0 m³/s
质量容差	1e-08 m³/s
能量残差	1.148e-09 m
能量容差	1e-05 m
迭代次数	5

模型假设

- 稳态不可压单相液态水
- 节点压力采用表压，H=z+p/(rho*g)
- 不含水锤、CFD 或时间状态

问题列表

级别	对象	说明
警告	用户末端 2	末端流量低于设计需求，固定工况不满足

支路结果

设备 ID	名称	类型	流量 [m³/h]	质量流量 [kg/h]	流速 [m/s]	测压水头差 [m]	水力功率 [W]	估算输入功率 [W]
P1	P1	pipe	31.17815	31123.08	1.722971	3.126067	—	—
P2	P2	pipe	13.6786	13654.44	1.145045	2.382101	—	—
P3	P3	pipe	13.6786	13654.44	1.145045	2.382101	—	—
P4	P4	pipe	31.17815	31123.08	1.722971	3.126067	—	—
PU1	循环泵 · 虚构示例	pump	31.17815	31123.08	—	-18.49893	1568.366	—
V1	调节阀 1	valve	17.49955	17468.64	—	0.4322102	—	—
T1	用户末端 1	terminal	17.49955	17468.64	—	11.81459	—	—
V2	调节阀 2	valve	13.6786	13654.44	—	0.2640735	—	—
T2	用户末端 2	terminal	13.6786	13654.44	—	7.218523	—	—

无效率数据时不提供输入功率；水力功率不等于整机电功率。压力水头不等于总水头。

完整输入（SI）


```
{
  "nodes": [
    {
      "id": "s0",
      "name": "泵出口",
      "elevation": 0,
      "demand": 0,
      "head": null
    },
    {
      "id": "s1",
      "name": "供水分配",
      "elevation": 0,
      "demand": 0,
      "head": null
    },
    {
      "id": "s2",
      "name": "远端供水",
      "elevation": 0,
      "demand": 0,
      "head": null
    },
    {
      "id": "t1",
      "name": "末端一入口",
      "elevation": 0,
      "demand": 0,
      "head": null
    },
    {
```

```
    "id": "t2",
    "name": "末端二入口",
    "elevation": 0,
    "demand": 0,
    "head": null
  },
  {
    "id": "r0",
    "name": "定压点 / 泵入口",
    "elevation": 0,
    "demand": 0,
    "head": 30
  },
  {
    "id": "r1",
    "name": "回水汇合",
    "elevation": 0,
    "demand": 0,
    "head": null
  },
  {
    "id": "r2",
    "name": "远端回水",
    "elevation": 0,
    "demand": 0,
    "head": null
  }
],
"edges": [
  {
    "id": "P1",
    "name": "P1",
    "kind": "pipe",
    "source": "s0",
    "target": "s1",
    "length": 75,
    "diameter": 0.08,
    "roughness": 5e-05,
    "zeta": 2,
    "local_mode": "zeta"
  },
  {
    "id": "P2",
    "name": "P2",
    "kind": "pipe",
    "source": "s1",
    "target": "s2",
    "length": 100,
    "diameter": 0.065,
    "roughness": 5e-05,
    "zeta": 2,
    "local_mode": "zeta"
  },
  {
    "id": "P3",
    "name": "P3",
    "kind": "pipe",
    "source": "r2",
    "target": "r1",
    "length": 100,
    "diameter": 0.065,
    "roughness": 5e-05,
    "zeta": 2,
    "local_mode": "zeta"
  },
```

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{
  "id": "P4",
  "name": "P4",
  "kind": "pipe",
  "source": "r1",
  "target": "r0",
  "length": 75,
  "diameter": 0.08,
  "roughness": 5e-05,
  "zeta": 2,
  "local_mode": "zeta"
},
{
  "id": "PU1",
  "name": "循环泵 · 虚构示例",
  "kind": "pump",
  "source": "r0",
  "target": "s0",
  "curve": [
    32,
    0,
    -180000
  ],
  "speed": 1,
  "parallel": 1,
  "check_valve": true,
  "valid_flow": [
    0,
    0.013
  ],
  "data_source": "人工构造，非厂家数据"
},
{
  "id": "V1",
  "name": "调节阀 1",
  "kind": "valve",
  "source": "s1",
  "target": "t1",
  "opening": 1,
  "kv": 85,
  "characteristic": "linear"
},
{
  "id": "T1",
  "name": "用户末端 1",
  "kind": "terminal",
  "source": "t1",
  "target": "r1",
  "design_head": 8,
  "design_flow": 0.004
},
{
  "id": "V2",
  "name": "调节阀 2",
  "kind": "valve",
  "source": "s2",
  "target": "t2",
  "opening": 1,
  "kv": 85,
  "characteristic": "linear"
},
{
  "id": "T2",
  "name": "用户末端 2",
  "kind": "terminal",
```

```
      "source": "t2",
      "target": "r2",
      "design_head": 8,
      "design_flow": 0.004
    }
  ],
  "fluid": {
    "temperature": 20,
    "atmospheric_pressure": 101325
  }
}
```